Question	Answer	Marks	Guidance
1	**Defining the problem**		Annotate with P.
	Vary *r* and measure *c* **or** *r* is the independent variable and *c* is the dependent variable	**1**	This mark could be awarded having read the candidate's script all the way through. **Not** *r* or *c* constant.
	Keep *v* constant	**1**	**Allow** same for constant **Allow** speed or v^2 for *v* **Ignore** keep distance ball falls constant see D2
	Methods of data collection		Annotate with M0.
	Labelled diagram of workable experiment including: • cardboard on springs • springs on bench • rule (to measure *c*) clamped vertically near to cardboard • stand on bench • two labels from bench, spring(s), cardboard, clamp / stand, rule	**1**	**Note** D2, D5, D7 **Allow** ruler for rule
	Method to determine *c*, e.g. measure initial height (of cardboard / spring) and new height **and** *c* = original height – new height	**1**	**Allow** difference in heights or new height – original height
	Method to determine *r* e.g. use micrometer / calipers to measure diameter *d* **and** $r = d / 2$	**1**	**Not** rule(r) **Allow** rule(r) **and** use of set squares / blocks
	Measure height (*h*) ball falls (from release point to cardboard) with rule (to determine *v*)	**1**	**Allow** ruler for rule **Not** micrometer / calipers

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Question	Answer	Marks	Guidance
1	**Method of Analysis**		Annotate with ✓.
	Plot a graph of $\frac{1}{c^2}$ against $\frac{1}{r^3}$ **or** plot a graph of $\frac{1}{r^3}$ against $\frac{1}{c^2}$ or equivalent. Do not accept logarithms.	**1**	**Note** drawn axes take priority.
	$\frac{1}{c^2}$ against $\frac{1}{r^3}$: $\alpha = \pi\rho v^2 \times$ gradient $\frac{1}{r^3}$ against $\frac{1}{c^2}$: $\alpha = \frac{\pi\rho v^2}{\text{gradient}}$	**1**	**Note** Must be consistent with drawn graph.
	$\frac{1}{c^2}$ against $\frac{1}{r^3}$: $Z = y\text{-intercept} \times v^2$ $\frac{1}{r^3}$ against $\frac{1}{c^2}$: $Z = -\frac{\alpha \times y\text{-intercept}}{\pi\rho}$ **or** $Z = -\frac{v^2 \times y\text{-intercept}}{\text{gradient}}$	**1**	**Note** Must be consistent with drawn graph.

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Question	Answer	Marks	Guidance
1	**Additional detail including safety considerations**	**6**	Annotate with numbered tick
	D1 Precaution linked <u>to stop ball</u> (leaving bench), e.g. screens around apparatus to stop the ball or cushions / sand tray on bench to stop the ball		**Ignore** goggles **Ignore** reference to floor **Ignore** collisions with equipment / people
	D2 Method to keep v constant, by keeping distance ball falls constant, e.g. electromagnet <u>fixed</u> between the springs to release (steel) ball		**Note** from diagram **Allow** <u>fixed</u> release point between springs e.g. clamp / rule fixed
	D3 Method to determine v using height h ball <u>falls</u> to cardboard e.g. $v = \sqrt{2gh}$ or $v^2 = 2gh$		**Ignore** timing methods **Note** h must be defined **Note** must use g
	D4 Measure mass m of ball using a balance **and** $\rho = \frac{m}{\frac{4}{3}\pi r^3}$ or $\rho = \frac{6m}{\pi d^3}$ **or** $\rho = \frac{m}{V}$ **and** $V = \frac{4}{3}\pi r^3$ or $\frac{1}{6}\pi d^3$		**Ignore** displacement methods **Allow** scale<u>s</u> **Ignore** scale
	D5 Description of method to ensure stability of the cardboard e.g. tape / adhesive putty at end of the spring(s) **or** position springs at the edges / corners of the cardboard or symmetrically		**Allow** from diagram **Allow** glue at top of springs **Ignore** clamped springs **Allow** springs fixed to base or rods clamped to bench
	D6 Method to ensure that c is the same for all springs, e.g. ball needs to hit centre of cardboard **or** use springs each with the <u>same spring constant</u>		**Ignore** symmetrical (D5) **Ignore** k is constant **Ignore** springs are the same
	D7 Description of method to determine accurate value of h and/or c e.g. measure from the bottom of the ball to the top of the cardboard **or** use of plumb line / set square to ensure rule is vertical		**Allow** on diagram **Note** reason required

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Question	Answer	Marks	Guidance
1	D8 Use springs with small value of spring constant / longer springs / springs with more turns / larger h to increase c **or** Use of video and slow-motion playback / frame by frame (with rule in view) to observe (minimum) new height (of spring/cardboard)		**Ignore** reference to pointer since there is the edge of the cardboard **Allow** determine c for new height
	D9 Repeat determination of c for the same value of r **and** average c		**Allow** same ball or r or d **Allow** each for same
	D10 Relationship valid if a straight line is produced (with y-intercept = $\frac{Z}{v^2}$). Do not accept through the origin.		**Note** Must be consistent with drawn graph. **Ignore** 'data' / 'graph' / 'experiment' / 'results' / 'correlation' valid.

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